VISUAL SPEAKER NOTES • CONTENT-DRIVEN EDITION

Fronting the Azure AI Stack with Azure API Management

Fronting the Azure AI Stack with Azure API Management | Udaiappa Ramachandran

Udaiappa Ramachandran • 2026-09-20 • 1:17:57

- 31 substantive visual(s), selected by content
- Repeated builds, presenter-only shots, and filler excluded
- Output length expands or contracts with the source material
- Speaker claims and caveats belong in the notes

https://www.youtube.com/watch?v=uGelt4MIK-M

The Problem: Islands of Intelligence



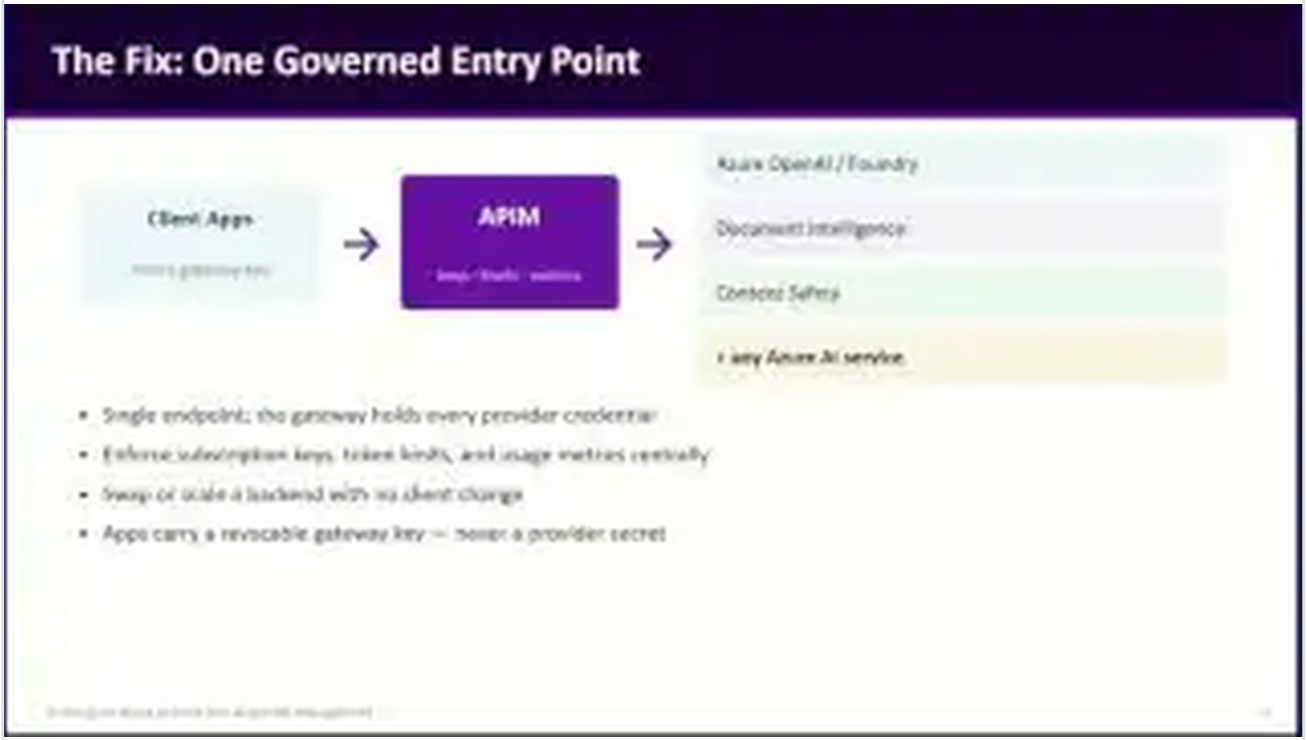
SPEAKER NOTES

- Direct integrations force every application to store provider endpoints and credentials and repeat wiring for Foundry, Document Intelligence, and Content Safety.
- The result is key sprawl, duplicated integration, shared quota contention, and no central view of token usage or cost by client or team.
- Key Vault protects secrets but does not remove duplicated service wiring, rotation propagation, or client-to-provider coupling.

WHY IT MATTERS

Direct connections multiply operational and security work as applications and AI services grow.

The Fix: One Governed Entry Point



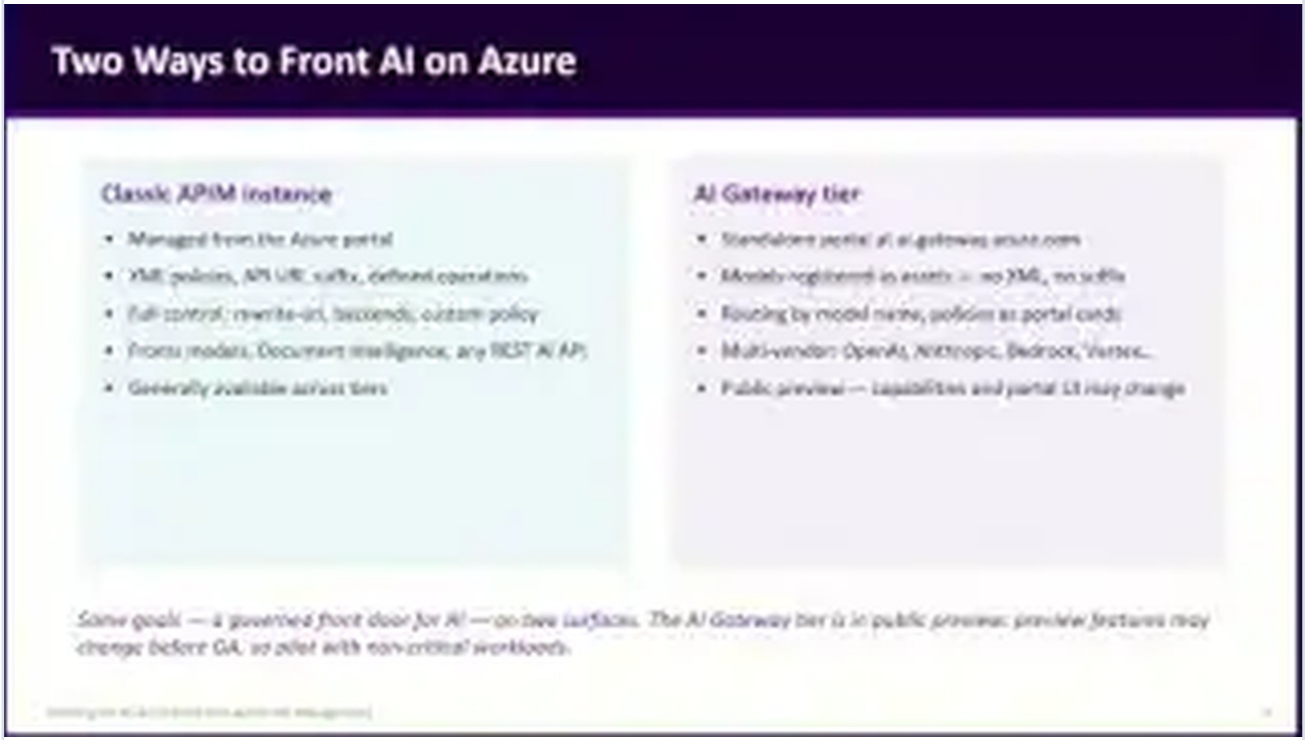
SPEAKER NOTES

- Change the request path to client → APIM → AI provider so all applications enter through one managed gateway.
- Give every application its own APIM subscription key for independent revocation, rotation, quotas, and usage attribution.
- Centralize token limits, caching, monitoring, and provider credentials in APIM; the gateway itself must still be scaled and monitored.

WHY IT MATTERS

The gateway separates client access from provider secrets and creates one enforceable policy layer.

Two Ways to Front AI on Azure



SPEAKER NOTES

- Classic APIM provides mature XML policies, explicit operations, backend definitions, URL rewriting, and REST-level control.
- The newer AI Gateway registers models as assets and routes by model or deployment name with less manual configuration.
- The speaker treats the AI Gateway tier as preview technology and recommends classic APIM for critical production use.

WHY IT MATTERS

Choose the mature gateway for control; use the simplified AI surface only when preview risk is acceptable.

How a Request Flows

The key to every gateway error



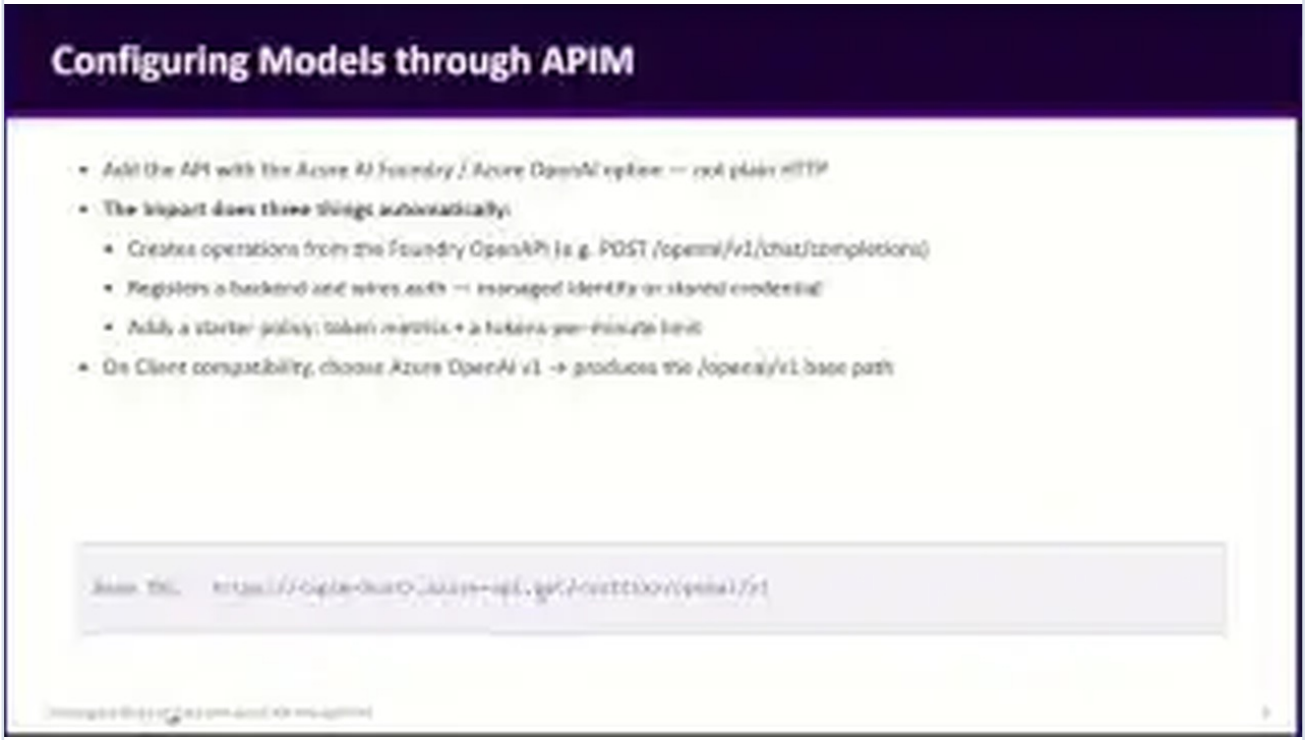
SPEAKER NOTES

- APIM first matches the host, API suffix, and operation template; an unmatched request returns 404 before policies execute.
- Only after a match do inbound policies select the backend, rewrite the URI, enforce limits, and forward the request.
- A policy cannot repair a route that never matched, so begin gateway 404 diagnosis with suffixes and operation definitions.

WHY IT MATTERS

Knowing the failing stage prevents debugging code or policies that never received the request.

Configuring Models through APIM



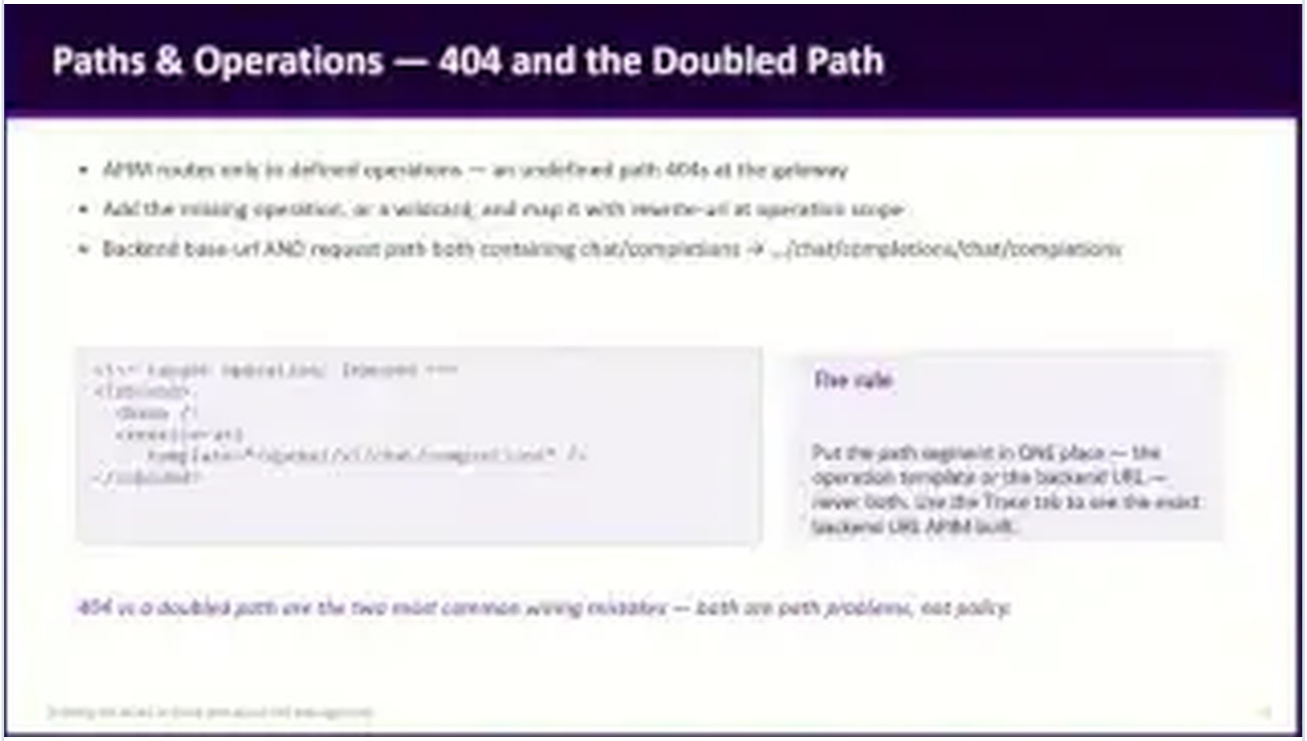
SPEAKER NOTES

- Import Foundry or Azure OpenAI with the provider-specific option rather than a plain HTTP API.
- The importer generates operations, registers the backend, configures managed identity or stored credentials, and adds starter token policies.
- For OpenAI-compatible clients, the gateway base URL ends in /openai/v1; older API-version URLs do not follow this pattern.

WHY IT MATTERS

Purpose-built import supplies routing and authentication plumbing that is easy to misconfigure manually.

Paths & Operations: 404 and the Doubled Path



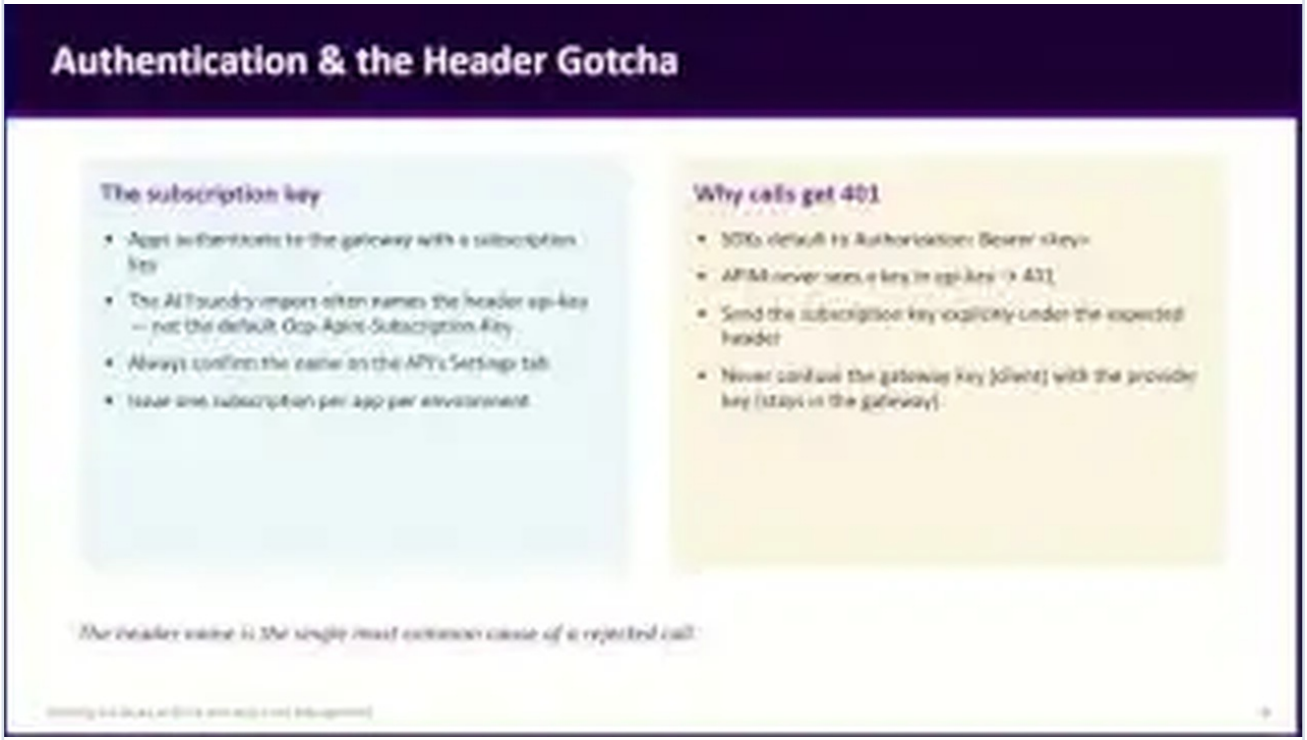
SPEAKER NOTES

- Undefined operations are rejected at APIM with 404 and never reach the backend.
- Add the named operation or use a wildcard temporarily during discovery, then replace it with explicit production routes.
- Keep every path segment in one place; repeating /openai/v1 in suffixes, operations, or rewrites creates doubled backend paths.

WHY IT MATTERS

Explicit path ownership makes gateway routing deterministic and 404 failures easier to isolate.

Authentication & the Header Gotcha



SPEAKER NOTES

- Create a separate APIM subscription for each application instead of sharing one key across all clients.
- Foundry imports may expect api-key while standard APIM often uses Ocp-Apim-Subscription-Key; verify the configured header.
- An SDK may place its key in Authorization: Bearer, which fails when APIM expects a subscription header; clients and settings must agree.

WHY IT MATTERS

Keep the client gateway credential distinct from the provider credential that APIM uses on the backend.

CORS for Browser Clients



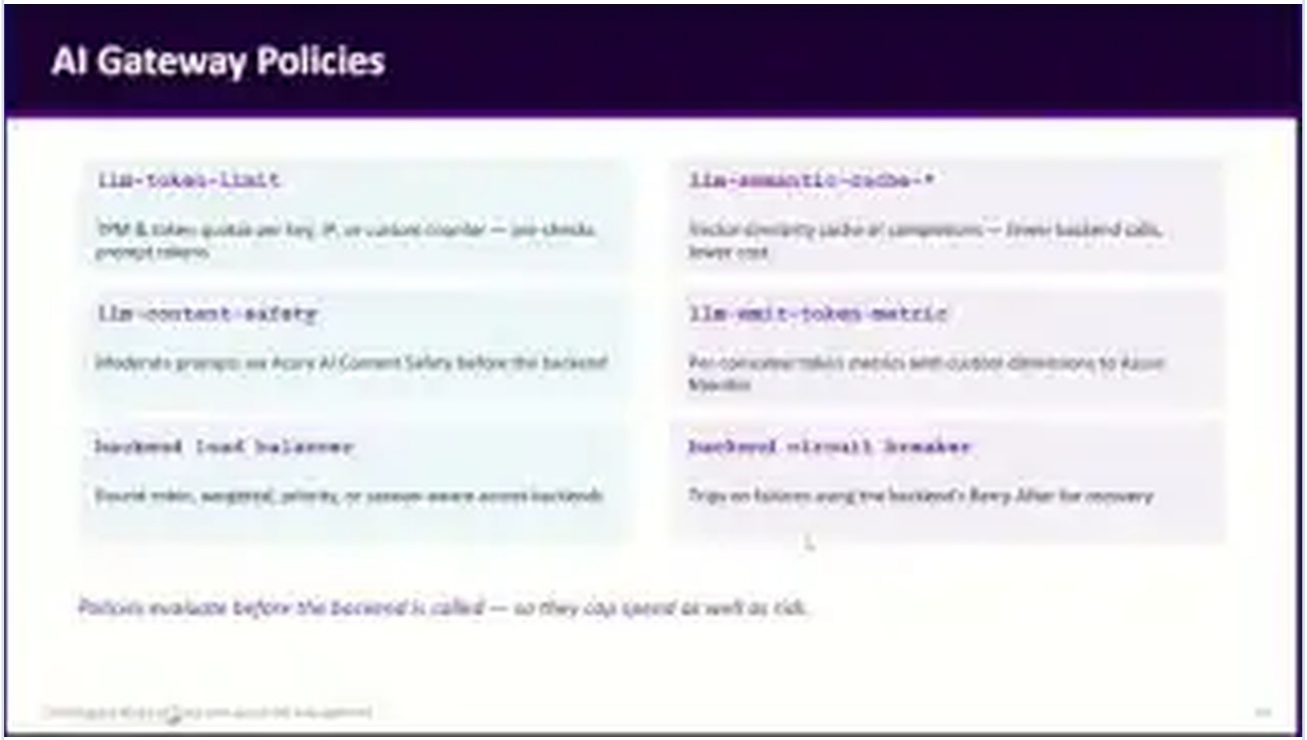
SPEAKER NOTES

- Browsers send OPTIONS preflight requests; APIM's CORS policy can answer them without forwarding to the model backend.
- Allow only the origins, methods, and headers the application needs, such as OPTIONS, POST, and GET.
- Wildcards are supported but the speaker advises against them for production browser applications.

WHY IT MATTERS

Gateway CORS enables browser clients without exposing unnecessarily broad cross-origin access.

AI Gateway Policies



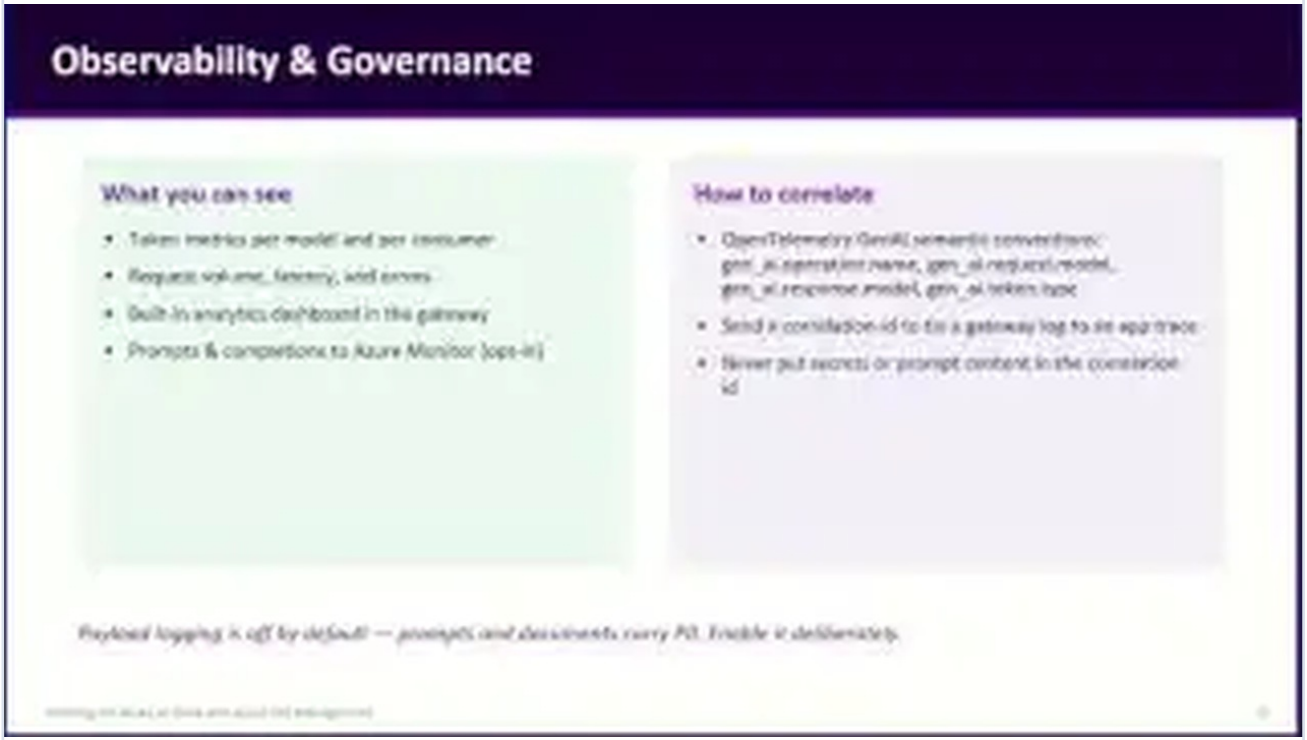
SPEAKER NOTES

- Token-limit policies reject oversized prompts before they consume model capacity or cost.
- Semantic caching can reuse responses for equivalent intent, while content safety and token metrics centralize governance.
- Backend balancing and circuit breaking improve resilience; semantic caching adds little when every request has a distinct intent.

WHY IT MATTERS

One gateway policy layer centralizes cost, safety, performance, and resilience controls.

Observability & Governance



SPEAKER NOTES

- APIM exposes token use by model and consumer together with request volume, latency, and error metrics.
- OpenTelemetry using GenAI conventions can feed the organization's existing monitoring pipeline.
- Correlation IDs isolate one slow or failed request; never log secrets or sensitive prompt content in captured headers.

WHY IT MATTERS

Per-request evidence turns aggregate AI usage into information operations and finance can act on.

Naming Is a Public Contract



SPEAKER NOTES

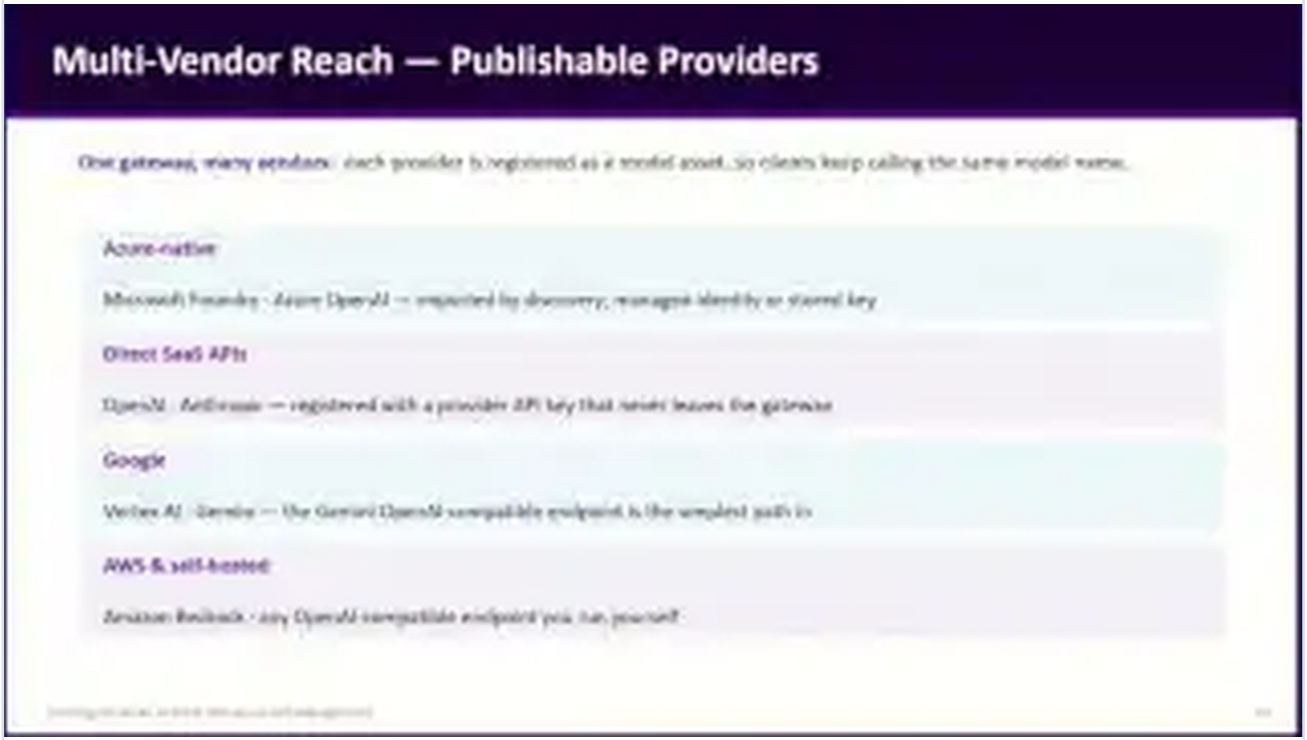
- The APIM gateway name becomes part of its URL and cannot be renamed after creation.
- Use a durable organization convention based on workload, environment, and region rather than one customer or provider.
- Vendor-neutral model names let clients keep one contract while APIM changes the backend; custom DNS adds portability but also cost and configuration.

WHY IT MATTERS

Gateway and model names become durable client dependencies, so naming decisions constrain future change.

Multi-Vendor Reach

Publishable providers behind one contract



SPEAKER NOTES

- Azure Foundry and Azure OpenAI can use discovery plus managed identity or stored keys.
- Direct SaaS providers such as OpenAI and Anthropic keep their provider keys at the gateway rather than in applications.
- Gemini, Bedrock, self-hosted models, and OpenAI-compatible endpoints can sit behind the same client contract, but provider protocols still differ.

WHY IT MATTERS

A stable gateway contract allows provider substitution without redistributing backend credentials to clients.

Backend Authentication Caveats by Vendor



SPEAKER NOTES

- OpenAI and Gemini AI Studio use static keys; Anthropic also requires its version header unless a compatible route is used.
- Google Vertex credentials expire, while Bedrock SigV4 signs the method, path, headers, body, and timestamp on every request.
- Bedrock may need an intermediary signing proxy; the speaker notes that this path was based mainly on documentation rather than deep hands-on use.

WHY IT MATTERS

Provider authentication determines whether APIM can call the service directly or needs a compatible proxy.

Document Intelligence through APIM



SPEAKER NOTES

- Document Intelligence extracts text from PDFs, images, Word files, and forms before a RAG pipeline chunks and embeds it.
- The initial analysis call is asynchronous and returns 202 Accepted; the client polls until the extracted result is ready.
- Unlike chat completion, this is not one request-response exchange and requires durable operation tracking.

WHY IT MATTERS

The same governed gateway can protect the document-ingestion side of a RAG architecture.

The Operation-Location Rewrite



SPEAKER NOTES

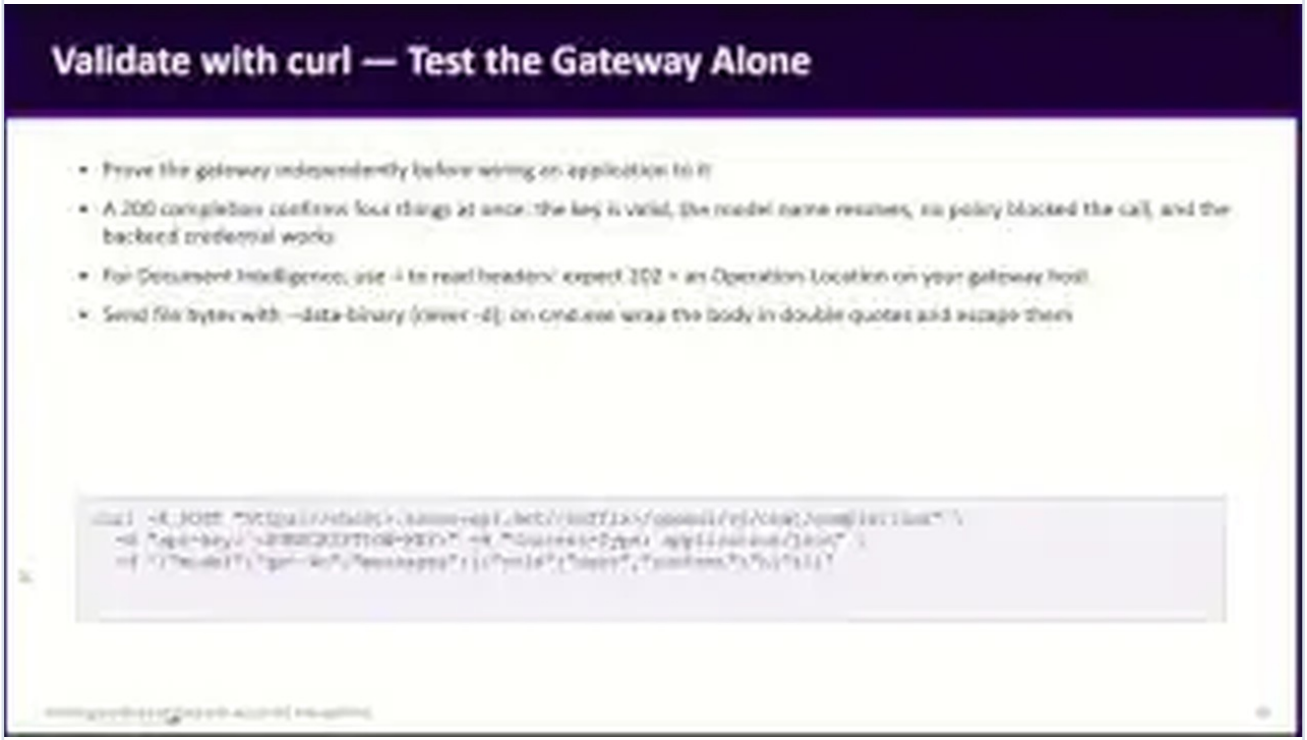
- A 202 response includes Operation-Location, which the client uses for status and result polling.
- The backend generates that URL with its own hostname, so APIM rewrites it to the gateway hostname before returning it.
- Incorrect rewriting can bypass APIM or make polling fail; a direct guarded string replacement is the simplest mapping.

WHY IT MATTERS

Rewriting preserves asynchronous polling without exposing or coupling the client to the provider host.

Validate with curl

Test the gateway alone



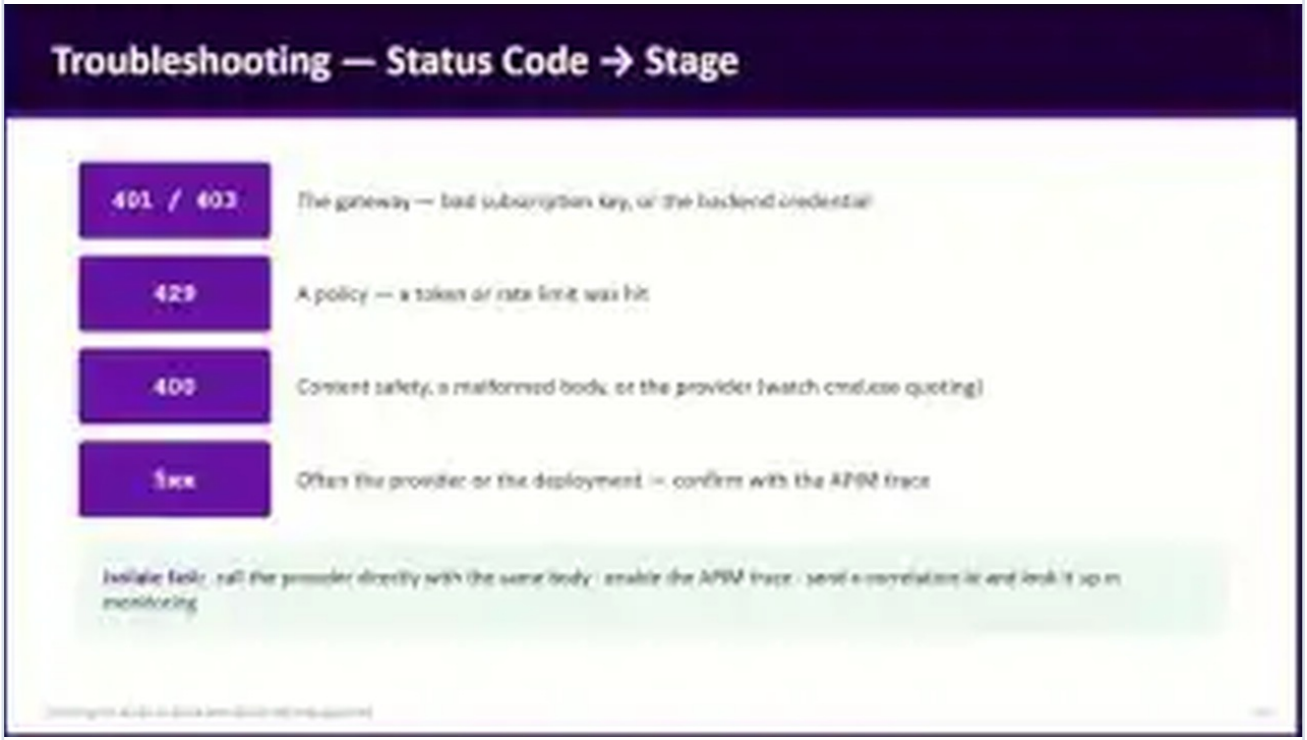
SPEAKER NOTES

- Start with APIM's test blade and trace, then reproduce the same method, headers, key, model, and body with curl.
- Compare environments to expose allow-list or network differences; if curl works, focus on application code next.
- Shell quoting differs across Command Prompt, PowerShell, and Unix shells and can itself produce misleading 400 errors.

WHY IT MATTERS

A minimal curl request separates gateway, credential, payload, network, and client-code failures.

Troubleshooting: Status Code → Stage



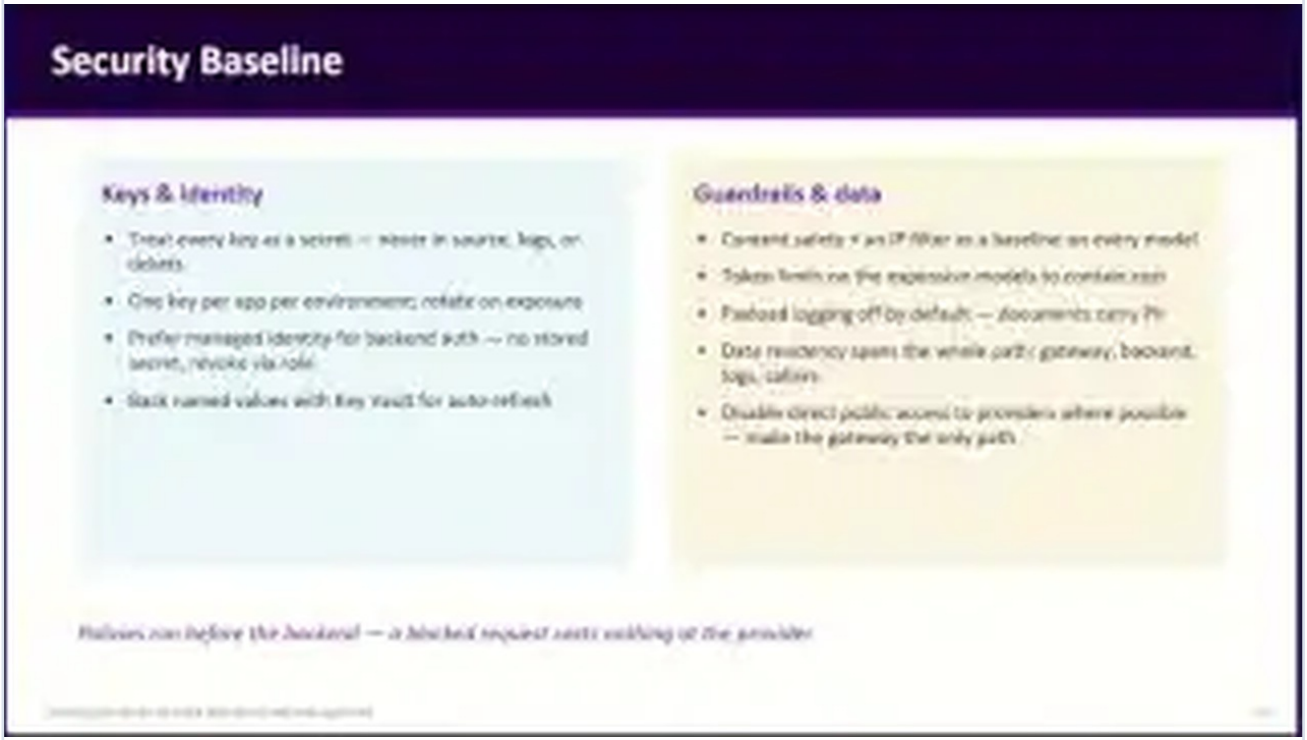
SPEAKER NOTES

- For 401/403, determine whether the client subscription key or APIM's backend credential failed.
- For 429, distinguish an APIM token policy from provider throttling; 400 usually indicates malformed input or safety rejection.
- A 5xx means the request progressed farther; replay it against the provider and carry a correlation ID through the path.

WHY IT MATTERS

Status codes identify the likely failing stage and prevent debugging components that never handled the request.

Security Baseline



SPEAKER NOTES

- Keep keys out of source, logs, and tickets; issue one key per application and environment and rotate after exposure.
- Prefer managed identity and RBAC; store remaining provider secrets in Key Vault and refresh rotated values.
- Apply safety, filtering, token limits, logging controls, and residency policies, then disable direct public provider access where possible.

WHY IT MATTERS

Gateway governance is enforceable only when clients cannot bypass APIM and call providers directly.

Importing an AI API into APIM



SPEAKER NOTES

- Choose the Microsoft Foundry or Azure OpenAI source so APIM creates provider operations and backend plumbing.
- The wizard configures key-based or managed-identity backend authentication and starter governance policies.
- A new APIM instance can take roughly 30 minutes to activate; automation should poll readiness rather than assume immediate availability.

WHY IT MATTERS

Provider-specific import avoids rebuilding operations, authentication, and policies by hand.

Policy XML as Versioned Configuration



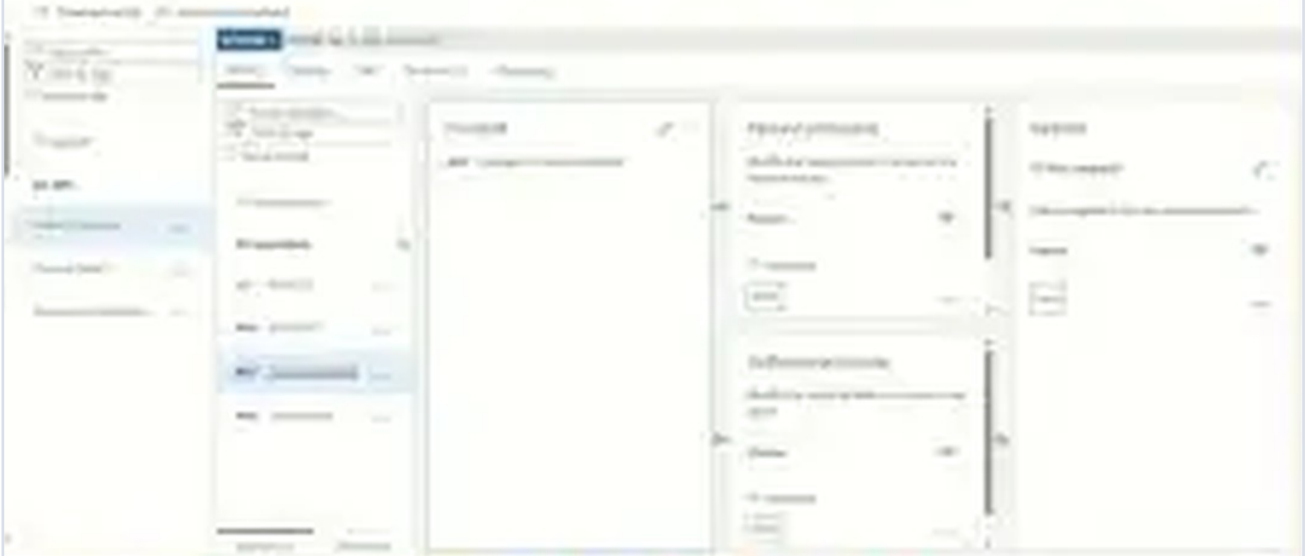
SPEAKER NOTES

- The sample repository keeps separate XML policies for Foundry, chat completion, Content Safety, and Document Intelligence.
- Automation imports those files, while the portal editor provides a direct review and troubleshooting surface.
- Shared behavior belongs at All operations; operation-specific policy handles narrower transformations.

WHY IT MATTERS

Versioned XML makes production gateway behavior repeatable and reviewable instead of portal-only state.

The APIM Operation Pipeline



SPEAKER NOTES

- Operation matching happens before the request enters inbound, backend, and outbound processing.
- All-operations policy runs first, followed by operation-specific behavior such as a Chat Completion request rewrite.
- Duplicating a path segment across suffixes, operations, and rewrites creates doubled URLs.

WHY IT MATTERS

The visual pipeline separates routing from policy execution and makes failures easier to localize.

APIM Test Console



SPEAKER NOTES

- Select an operation, open Test, and provide the subscription header and raw JSON body.
- Execute the request to inspect the response and trace, then reproduce it with an equivalent curl command.
- The portal may provide only a generic body, so the correct operation payload may need to be entered manually.

WHY IT MATTERS

A successful portal test proves the gateway path before client application behavior is investigated.

Create Chat Completion Operation



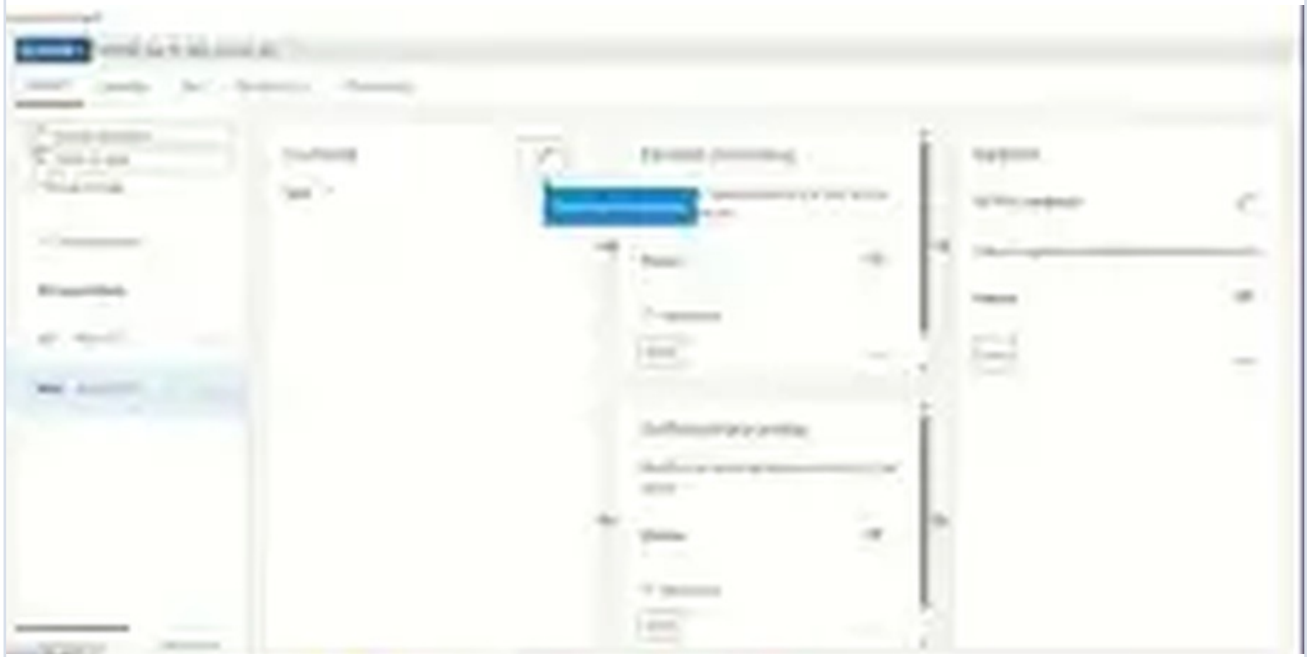
SPEAKER NOTES

- Chat Completion is exposed separately from embeddings and the Responses API.
- Apply model-specific request normalization at this operation—for example, translating max_tokens to max_completion_tokens.
- Do not apply chat-body rewrites to embeddings, which return floating-point vector arrays.

WHY IT MATTERS

Operation-level policy confines compatibility fixes to the endpoint and model that require them.

Forward Request and Backend Selection



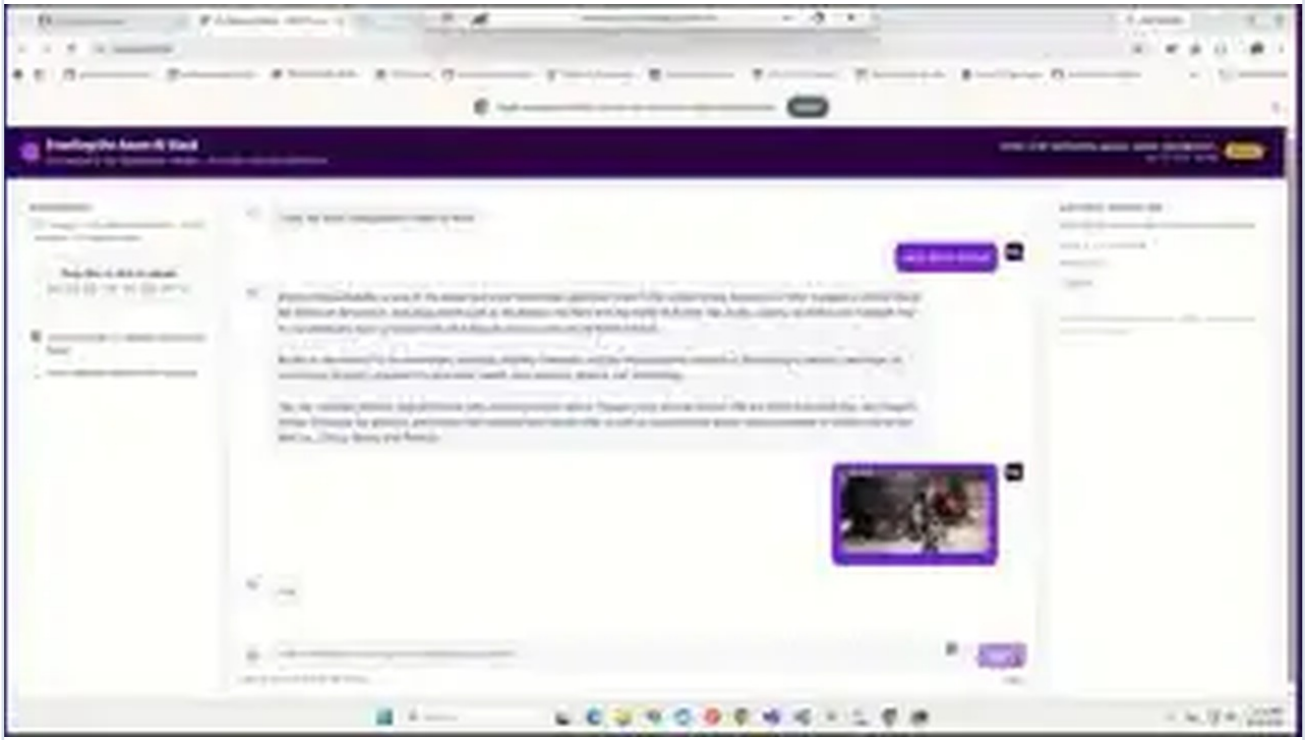
SPEAKER NOTES

- After a route matches, inbound policy can choose the backend, rewrite the URI, enforce controls, and forward the request.
- Applications replace a direct Foundry endpoint and key with the APIM endpoint and subscription key.
- Forward request is never reached for an undefined operation that already failed with a 404.

WHY IT MATTERS

Centralized forwarding lets clients gain governance without rewriting their core AI logic.

Direct-Service Baseline and Grounded Answer



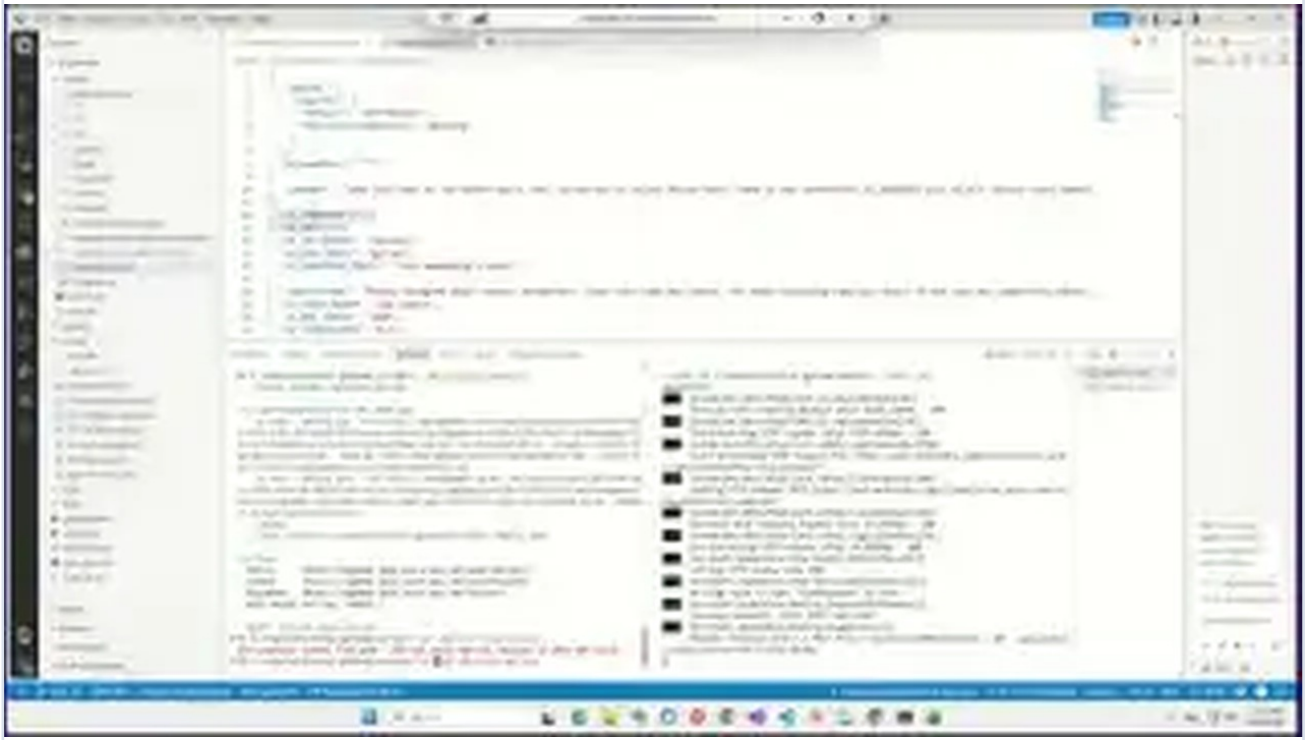
SPEAKER NOTES

- The baseline application calls chat, Content Safety, Document Intelligence, and embeddings directly against Azure services.
- A PowerPoint upload is extracted and chunked, then embeddings retrieve supporting context for the answer.
- The demo stores vectors in memory; production needs durable vector-capable storage.

WHY IT MATTERS

The direct-service baseline defines the behavior that must survive migration behind APIM.

Switch the Application from Foundry to APIM



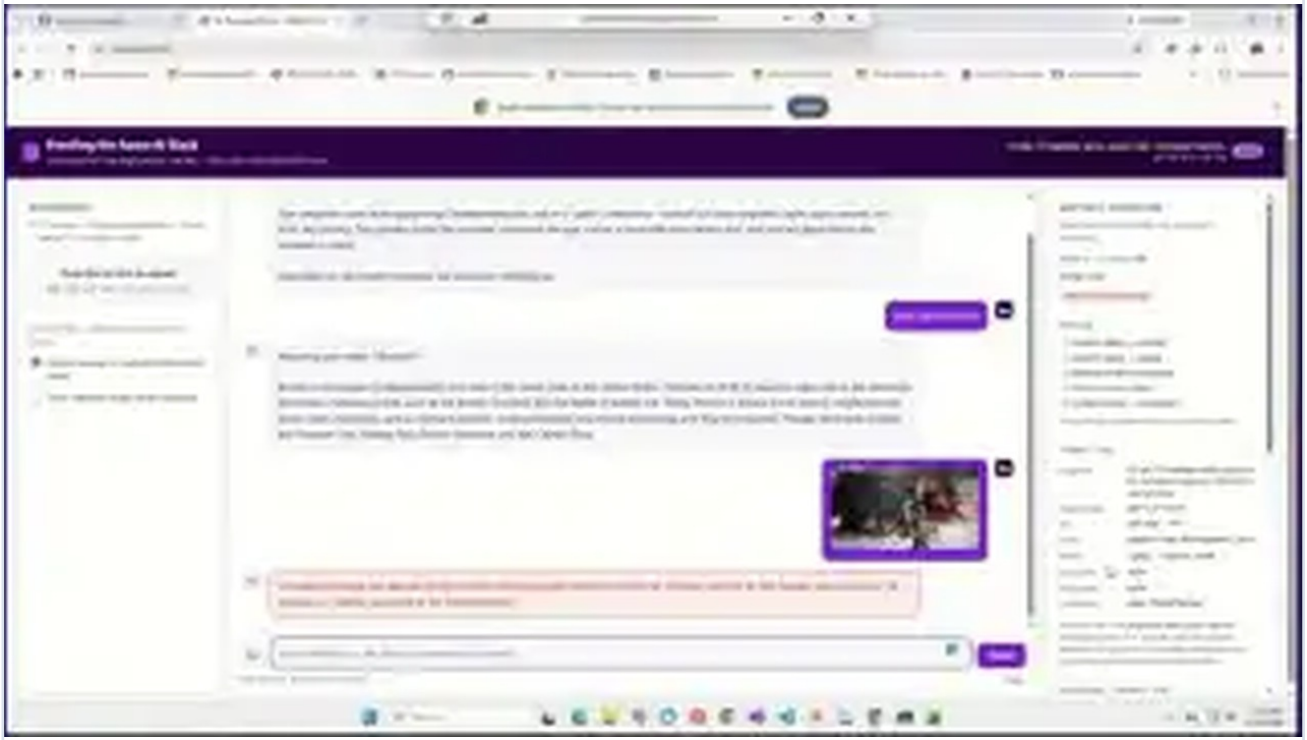
SPEAKER NOTES

- The environment script backs up the previous settings before changing the endpoint configuration.
- The OpenAI endpoint changes from direct Foundry access to APIM while the existing header carries the APIM subscription key.
- The application code is not rewritten; the same header becomes mandatory once APIM validates it.

WHY IT MATTERS

Moving behind the gateway is primarily a configuration change, which reduces migration risk.

Post-Migration Regression Test



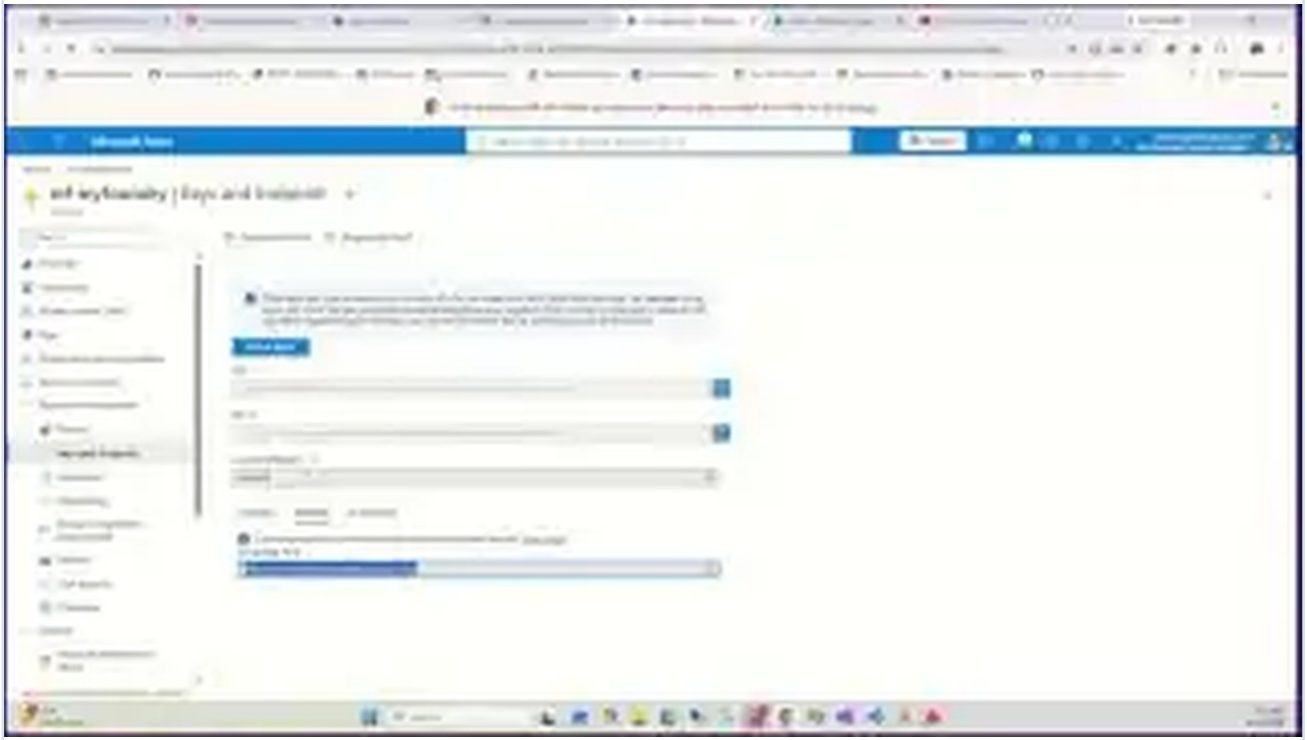
SPEAKER NOTES

- The presenter repeats chat, harmful-content, document-upload, and retrieval tests through APIM.
- Document extraction and retrieval still produce four matched chunks for the answer.
- This is functional validation rather than a controlled quality benchmark.

WHY IT MATTERS

The gateway adds governance while preserving the application's existing AI workflow.

APIM API and Policy Configuration



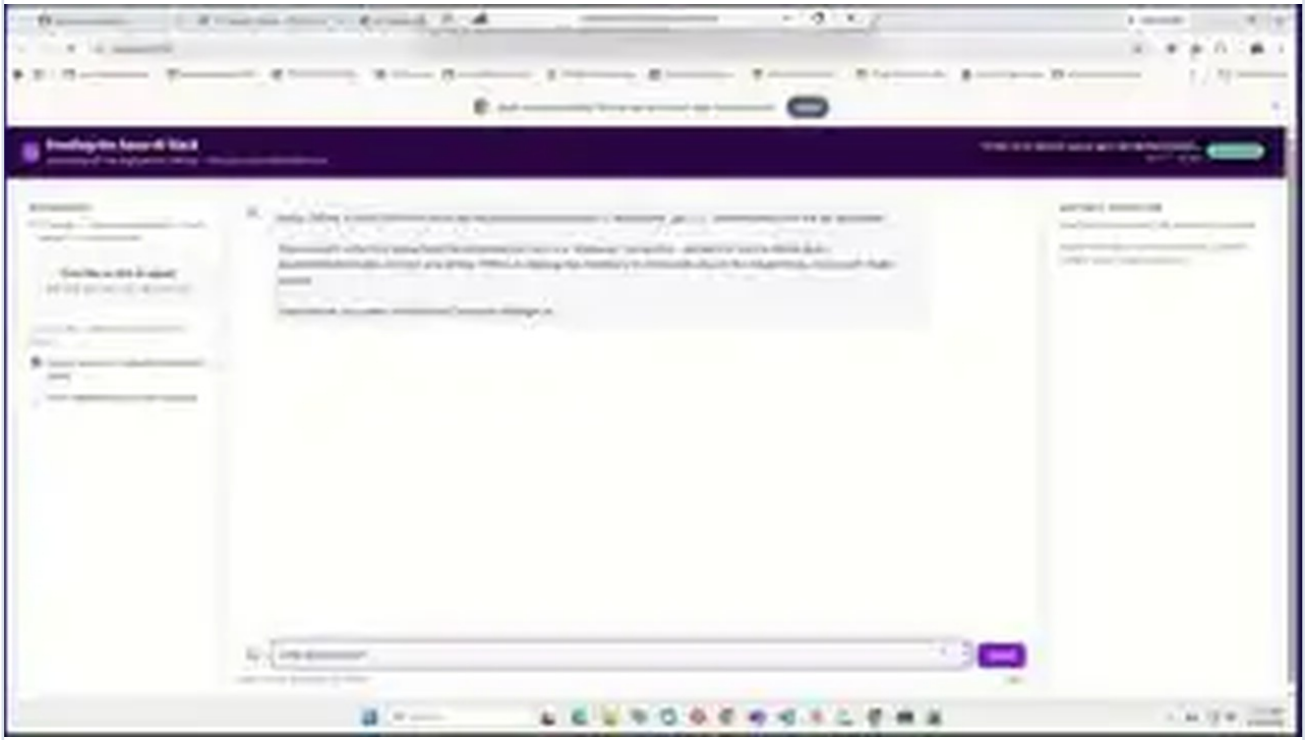
SPEAKER NOTES

- APIM contains separate interfaces for Foundry, Content Safety, and Document Intelligence plus shared all-operations policy.
- Operation-specific policy can override shared behavior, such as adapting max_tokens for one chat model.
- Settings and Test expose the endpoint, subscription-header configuration, and raw execution path.

WHY IT MATTERS

The gateway centralizes routing and request adaptation instead of duplicating them across applications.

AI Gateway Failure, Correction, and Preview Risk



SPEAKER NOTES

- The first call returns 404 because the configured model does not match an available deployment.
- After switching to the deployed model and correcting the endpoint, the request completes.
- The speaker reports inconsistent preview behavior, including failures inside the gateway's own test experience.

WHY IT MATTERS

Validate the model and endpoint in the gateway before debugging application code.

Monitoring Usage after the Tests



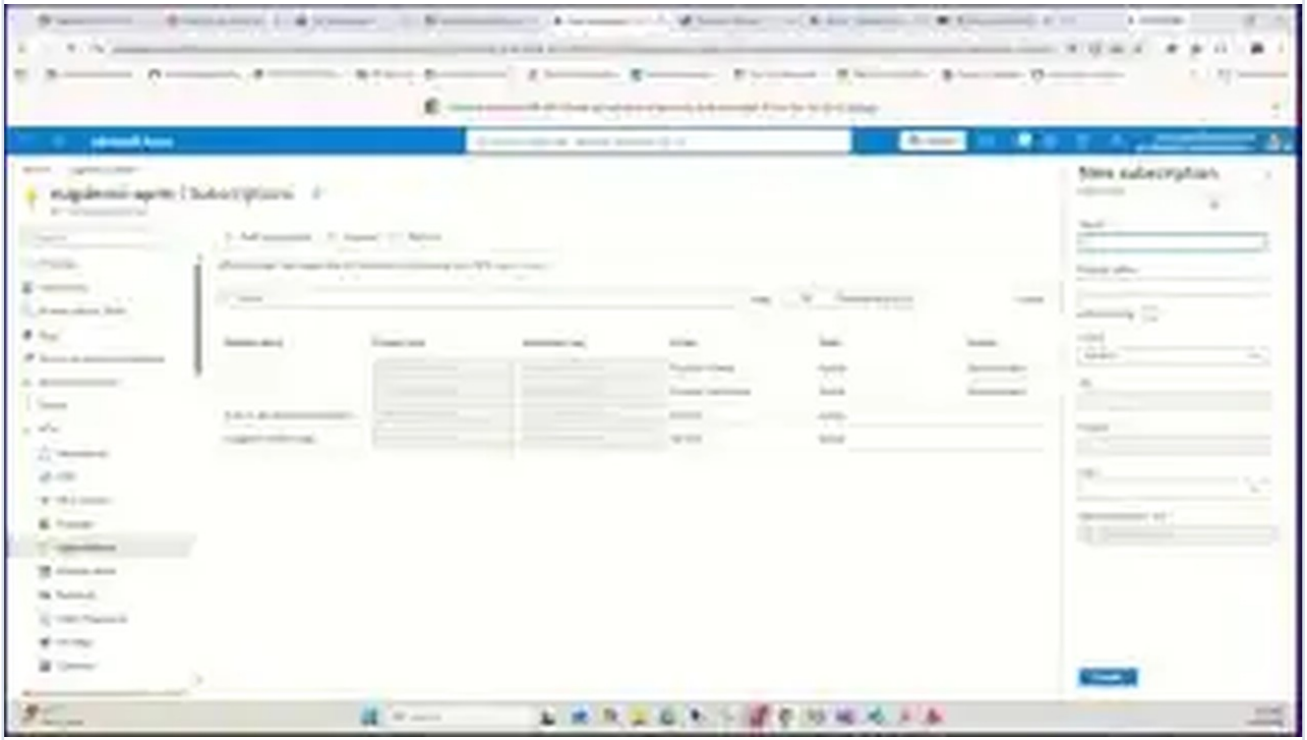
SPEAKER NOTES

- Monitoring shows activity after the application exercises the routed AI stack.
- Document Intelligence is billed by pages, while embeddings and language models use token-based accounting.
- Retrieved chunks contribute input tokens and the generated answer contributes output tokens; quoted costs are time- and model-specific.

WHY IT MATTERS

Centralized monitoring makes cost and request behavior visible after traffic moves behind the gateway.

Per-Client APIM Subscription Keys



SPEAKER NOTES

- The subscription view creates separate gateway credentials for individual client applications.
- Clients receive a revocable APIM key rather than the provider credential.
- Independent subscriptions improve issuance and revocation boundaries but do not replace secure client-side secret handling.

WHY IT MATTERS

Per-client subscriptions turn one shared gateway into an enforceable access boundary.